

QN 1 Output:

```
/mnt/Development/Programming/Collage/Sixth/Compiler main*
• > cd "/mnt/Development/Programming/Collage/Sixth/Compiler/" && g++ lab_1.cpp -o lab_1
Enter comment: // Keshav
It is a comment

/mnt/Development/Programming/Collage/Sixth/Compiler main* 6s
• > cd "/mnt/Development/Programming/Collage/Sixth/Compiler/" && g++ lab_1.cpp -o lab_1
Enter comment: Keshav
It is not a comment

/mnt/Development/Programming/Collage/Sixth/Compiler main*
• > cd "/mnt/Development/Programming/Collage/Sixth/Compiler/" && g++ lab_1.cpp -o lab_1
Enter comment: /*Keshav*/
It is a comment
```

QN 2 Output:

```
/mnt/Development/Programming/Collage/Sixth/Compiler main*
• > cd "/mnt/Development/Programming/Collage/Sixth/Compiler/" && g++
Enter a string: aaaaa
aaaaa is accepted under rule 'a*'

/mnt/Development/Programming/Collage/Sixth/Compiler main*
• > cd "/mnt/Development/Programming/Collage/Sixth/Compiler/" && g++
Enter a string: aaabb
aaabb is accepted under rule 'a*b+'

/mnt/Development/Programming/Collage/Sixth/Compiler main* 37s
• > cd "/mnt/Development/Programming/Collage/Sixth/Compiler/" && g++
Enter a string: abb
abb is accepted under rule 'abb'
```

QN 3 Output:

```
/mnt/Development/Programming/Collage/Sixth/Compiler main*
• > cd "/mnt/Development/Programming/Collage/Sixth/Compiler/" && g++ 1
Enter an identifier: area_111
Valid identifier

/mnt/Development/Programming/Collage/Sixth/Compiler main* 21s
• > cd "/mnt/Development/Programming/Collage/Sixth/Compiler/" && g++ 1
Enter an identifier: 1Keshav
Not a valid identifier

/mnt/Development/Programming/Collage/Sixth/Compiler main*
• > cd "/mnt/Development/Programming/Collage/Sixth/Compiler/" && g++ 1
Enter an identifier: _Keshav
Valid identifier
Focus folder in explorer (ctrl + click)

/mnt/Development/Programming/Collage/Sixth/Compiler main*
• > cd "/mnt/Development/Programming/Collage/Sixth/Compiler/" && g++ 1
Enter an identifier: Keshav@124
Not a valid identifier
```

QN 4 Output:

```
/mnt/Development/Programming/Collage/Sixth/Compiler main*
• > cd "/mnt/Development/Programming/Collage/Sixth/Compiler/" && g++
Enter code/tokens: Keshav_N,5,a,while,if,+,*
Keshav_N is identifier
5 is identifier
a is identifier
while is keyword
if is keyword
+ is operator
* is operator
',' is separator for this input stream
```

QN 5 Output:

```
/mnt/Development/Programming/Collage/Sixth/Compiler main*
• > cd "/mnt/Development/Programming/Collage/Sixth/Compiler/" && g++ ]
How many number of productions? : 5
Enter production Number 1 : S=L=R
Enter production Number 2 : S=R
Enter production Number 3 : L=*R
Enter production Number 4 : L=a
Enter production Number 5 : R=L

Find the FIRST of : S

FIRST(S) = { * a }
Press 'y' to continue : y

Find the FIRST of : L

FIRST(L) = { * a }
Press 'y' to continue : y

Find the FIRST of : R

FIRST(R) = { * a }
Press 'y' to continue : n
```

QN 6 Output:

```
/mnt/Development/Programming/Collage/Sixth/Compiler main*
• > cd "/mnt/Development/Programming/Collage/Sixth/Compiler/" && g++ lab_6.cpp
Enter the no. of productions: 5
Enter 5 productions
Production with multiple terms should be given as separate productions
R=aS
R=(R)S
S=+RS
S=aRS
S=*S
Find FOLLOW of --> R
FOLLOW(R) = { $ ) * + a }
Do you want to continue(Press 1 to continue....)? 1
Find FOLLOW of --> S
FOLLOW(S) = { $ ) * + a }
Do you want to continue(Press 1 to continue....)? 1
Find FOLLOW of --> aRS
```

QN 7 Output:

```
• > cd "/mnt/Development/Programming/Collage/Sixth/Compiler/" &&  
Enter the input string: i*i+i
```

Stack	Input
\$e	i*i+i\$
\$bt	i*i+i\$
\$bcf	i*i+i\$
\$bci	i*i+i\$
\$bc	*i+i\$
\$bcf*	*i+i\$
\$bcf	i+i\$
\$bci	i+i\$
\$bc	+i\$
\$b	+i\$
\$bt+	+i\$
\$bt	i\$
\$bcf	i\$
\$bci	i\$
\$bc	\$
\$b	\$
\$	\$

SUCCESS

QN 8 Output:

```
/mnt/Development/Programming/Collage/Sixth/Compiler main*
● > cd "/mnt/Development/Programming/Collage/Sixth/Compiler/" && g++
SHIFT REDUCE PARSER
GRAMMAR:
E -> E+E
E -> E/E
E -> E*E
E -> a | b

Enter the input symbol: a+a/a

Stack      Input Symbol  Action
-----
$          a+a/a$    --
$a         +a/a$     Shift a
$E         +a/a$     Reduce E->a
$E+       a/a$      Shift +
$E+a      /a$        Shift a
$E+E      /a$        Reduce E->a
$E        /a$        Reduce E->E+E
$E/       a$        Shift /
$E/a      $         Shift a
$E/E      $         Reduce E->a
$E        $         Reduce E->E/E
$E        $         ACCEPT
```

QN 9 Output:

```
/mnt/Development/Programming/Collage/Sixth/Compiler main*
● > cd "/mnt/Development/Programming/Collage/Sixth/Compiler/" && g++
INTERMEDIATE CODE GENERATION

Enter the Expression : w:a+b*c

The intermediate code:      Expression
Z := b * c                  w:a+Z
Y := a + Z                  w:Y
Y := ;
```